AEB Sensor Configuration File

1. Introduction

This document describes the sensor configuration and parameterization used for the Automatic Emergency Braking system.

2. Radar Sensor Configuration

Parameter	**Configured Value**
Sensor Type	Ground Truth Radar Sensor
Sensor Index	0
Sampling Mode	Continuous
Detection Direction	Front-facing
Primary Signals	DistX, VrelX

3. Camera Sensor Configuration

Parameter	**Configured Value**
Camera Type	Stereo Camera
Field of View (H)	50 deg
Field of View (V)	28 deg
Range	1–100 m
Resolution	1280 × 960
Baseline	0.12 m
Focal Length	2400 px

4. Camera Output Signals

Signal	**Description**
ObjID	Detected object ID
Type	Object type classification
Confidence	Detection confidence level

nVisPixels	Visible object pixel count

5. Sensor Fusion Configuration

Subsystem	Role
Radar	Primary longitudinal control
Camera	Object validation
Fusion Logic	Combined decision support

6. Threshold Parameters

Thresholds were selected to ensure stable braking response and collision avoidance behavior.

Parameter	Value
Safe Distance	3 m
TTC Threshold	1.5 s
Confidence Threshold	0.7
Target Speed	30 km/h

7. Conclusion

The configured sensor architecture provides reliable environmental perception and supports stable Automatic Emergency Braking operation.